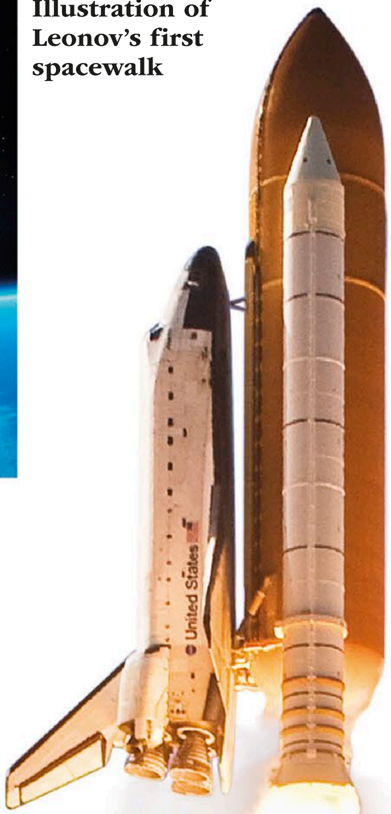
Illustration of Leonov's first spacewalk

SPACE WALKERS

Early astronauts wore flight suits and helmets for emergencies, but did not expect to leave their capsules. In March 1965, Voskhod 2 cosmonaut Alexei Leonov wore a specially designed suit to carry out the first walk in space. A few months later, astronaut Ed White made the first American spacewalk, steering himself with a gunlike personal thruster.

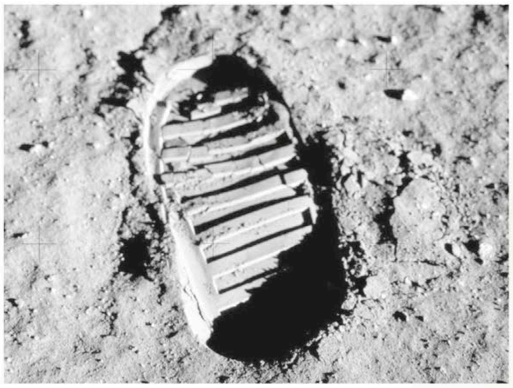


◂ LAUNCH OF SPACE SHUTTLE ATLANTIS ON STS-129 MISSION
One of five operational orbiters, Atlantis flew 33 missions, including this one to the International Space Station in 2009.

SPACEPLANE TO ORBIT

The Space Shuttle, flown by NASA from 1981 to 2011, took a totally new approach to spaceflight. Crews of up to seven flew aboard a planelike orbiter spacecraft with a large cabin and cargo hold. Its engines drew fuel from a huge tank during launch. On reaching orbit, the fuel tank fell away and the spacecraft continued on its journey. At the end of its mission, it glided back to Earth like a giant paper plane and could then be reused on other missions.

Buzz Aldrin's footprint on the moon, photographed on the Apollo 11 mission



MISSION TO THE MOON

The Apollo moon missions of the late 1960s used a complex three-part spacecraft. An orbiting command module carrying three astronauts was attached to a service module that carried supplies for missions of up to 12 days, while a lunar module carried two of the astronauts from the command module to the moon's surface and back. In July 1969, Apollo 11's Neil Armstrong and Buzz Aldrin became the first men to set foot on the moon.

Orion capsule

The crew module is 16.4ft (5m) in diameter and 10ft (3m) long.

FUTURE MISSIONS

Scheduled to fly in the 2020s, NASA's Orion spacecraft features an Apollo-like craft with a conical crew module attached to a cylindrical service module. However, Orion is far larger, and the reusable crew module will be able to carry crews of four to six on missions of up to 6 months in order to explore Mars and asteroids, conduct experiments, and service the International Space Station.